	C1 · Architecture and symbolic participation				C2 · Explanation and human validation		
Study	**Kautz**	**Strategy**	**Stage**	**Function**	**Explanation**	**Evaluation**	**HITL**
Entity and knowledge-graph alignment (n = 13)							
KnowAug (Li 2021)	IV	Logic	N1	Cand.	**Trace**	**Example**	No
SLHCat (Wang 2023)	IV	Struct.	N1	Cand.	None	None	No
CyberEA (Huang 2025)	IV	Logic	N1	Cand.	None	None	No
OntoEA (Xiang 2021)	V	Logic	N1	Constr.	None	None	No
Jiang et al. 2022	V	Logic	N1	Constr.	None	None	No
LLM4EA (Chen 2024)	IV	Logic	N1	Constr.	None	None	No
CIDER-LM (Kumar 2025)	IV	Logic	N1	Constr.	None	None	No
HSMGNN (Deng 2025)	V	Logic	N1	Constr.	None	None	No
PRASE (Qi 2021)	III	Logic	N2	Decision	**Score**	None	No
NALA (Xu 2024)	III	Logic	N2	Decision	**Native**	**Example**	No
NeuSymEA (Chen 2024)	V	Logic	N2	Decision	**Native**	**Proxy**	No
FLORA (Peng 2025)	II	Logic	N2	Decision	**Native**	**Example**	No
PhenoAlign (Yang 2025)	III	Logic	N2	Decision	None	None	No
Ontology matching and revision (n = 9)							
OntoConnect (Chakraborty 2021)	IV	Logic	N1	Cand.	None	None	No
BioHAN (Wang 2022)	IV	Logic	N1	Cand.	None	None	No
DAEOM (Wu 2020)	IV	Struct.	N1	Constr.	None	None	No
DeepFCA (Li 2020)	V	Logic	N1	Constr.	None	None	No
FCA-Map (Li 2021)	III	Logic	N2	Constr.	None	None	No
LogMap-ML (Chen 2021)	III	Logic	N2	Constr.	None	None	No
BERTMap (He 2021)	III	Logic	N2	Repair	None	None	No
Ji et al. 2023	II	Logic	N2	Repair	**Trace**	None	No
Lushnei et al. 2025	II	Logic	N2	Decision	None	None	No
Schema matching and data harmonization (n = 3)							
Mehryar et al. 2023	V	Logic	N1	Constr.	None	None	No
SMoG (Jeon 2025)	III	Struct.	N2	Cand.	**Trace**	**Proxy**	No
ReasonAlign (Zhou 2025)	II	Logic	N2	Decision	None	None	No
Entity resolution and record matching (n = 2)							
HMatch (Jradeh 2025)	III	Logic	N2	Expl.	**Native**	**Proxy**	No
DMatch (Deng 2022)	II	Logic	N2	Decision	**Native**	**Example**	No
Total (N = 27)	II **5** III **8** IV **8** V **6**	Logic **24** Struct. **3**	N1 **13** N2 **14**	Cand. **6** Constr. **10** Repair **2** Expl. **1** Decision **8**	Native **5** Trace **3** Score **1** None **18**	User **0** Example **4** Proxy **3** None **20**	Yes **0** No **27**